Chapter4 Lesson3 Atomic Structure

1. **(i)** Calculate the number of electrons which will together weigh one gram.
   (ii) Calculate the mass and charge of one mole of electrons.
2. **(i)** Calculate the total number of electrons present in one mole of methane.
   (ii) Find **(a)** the total number and **(b)** the total mass of neutrons in $7\ \text{mg}$ of 14C. (Assume that mass of a neutron $= 1.675 \times 10^{-27}\ \text{kg}$).
   (iii) Find (a) the total number and (b) the total mass of protons in $34\ \text{mg}$ of NH_3 at STP. Will the answer change if the temperature and pressure are changed?
3. How many neutrons and protons are there in the following nuclei?
   ${}^{13}_{6}\text{C}$, ${}^{16}_{8}\text{O}$, ${}^{12}_{24}\text{Mg}$, ${}^{56}_{26}\text{Fe}$, ${}^{88}_{38}\text{Sr}$
4. Write the complete symbol for the atom with the given atomic number (Z) and atomic mass (A)
   (i) $\text{Z} = 17$, $\text{A} = 35$.
   (ii) $\text{Z} = 92$, $\text{A} = 233$.
   (iii) $\text{Z} = 4$, $\text{A} = 9$.
5. Yellow light emitted from a sodium lamp has a wavelength (λ) of $580\ \text{nm}$. Calculate the frequency (ν) and wavenumber $(\bar{\upsilon})$ of the yellow light.
6. Find energy of each of the photons which
   (i) correspond to light of frequency $3\times10^{15}\ \text{Hz}$.
   (ii) have wavelength of $0.50\ \text{Å}$.
7. Calculate the wavelength, frequency and wavenumber of a light wave whose period is $2.0 \times 10^{-10}\ \text{s}$.
8. What is the number of photons of light with a wavelength of $4000\ \text{pm}$ that provide 1J of energy?
9. A photon of wavelength $4 \times 10^{-7}\ \text{m}$ strikes on metal surface, the work function of the metal being $2.13\ \text{eV}$. Calculate
   (i) the energy of the photon (eV), **(ii)** the kinetic energy of the emission, and (iii) the velocity of the photoelectron $(1\ \text{eV} = 1.6020 \times 10^{-19}\ \text{J})$.
10. Electromagnetic radiation of wavelength $242\ \text{nm}$ is just sufficient to ionise the sodium atom. Calculate the ionisation energy of sodium in kJ mol^{-1}.
11. A 25 watt bulb emits monochromatic yellow light of wavelength of $0.57\mu\text{m}$. Calculate the rate of emission of quanta per second.
12. Electrons are emitted with zero velocity from a metal surface when it is exposed to radiation of wavelength $6800\ \text{Å}$. Calculate threshold frequency (ν_0) and work function (W_0) of the metal.
13. What is the wavelength of light emitted when the electron in a hydrogen atom undergoes transition from an energy level with $\text{n} = 4$ to an energy level with $\text{n} = 2$?
14. How much energy is required to ionise a H atom if the electron occupies $\text{n} = 5$ orbit? Compare your answer with the ionization enthalpy of H atom (energy required to remove the electron from $\text{n} = 1$ orbit).

15. What is the maximum number of emission lines when the excited electron of a H atom in $n = 6$ drops to the ground state?

16. **(i)** The energy associated with the first orbit in the hydrogen atom is -2.18×10^{-18} J atom^{-1}. What is the energy associated with the fifth orbit?

(ii) Calculate the radius of Bohr's fifth orbit for hydrogen atom.

17. Calculate the wavenumber for the longest wavelength transition in the Balmer series of atomic hydrogen.

18. What is the energy in joules, required to shift the electron of the hydrogen atom from the first Bohr orbit to the fifth Bohr orbit and what is the wavelength of the light emitted when the electron returns to the ground state? The ground state electron energy is -2.18×10^{-11} ergs.

19. The electron energy in hydrogen atom is given by $E_n = (-2.18 \times 10^{-18})/n2$ J. Calculate the energy required to remove an electron completely from the $n = 2$ orbit. What is the longest wavelength of light in cm that can be used to cause this transition?

20. Calculate the wavelength of an electron moving with a velocity of 2.05×10^7 m s^{-1}.

21. The mass of an electron is 9.1×10^{-31} kg. If its K.E. is 3.0×10^{-25} J, calculate its wavelength.

22. Which of the following are isoelectronic species i.e., those having the same number of electrons?

23. Show that the circumference of the Bohr orbit for the hydrogen atom is an integral multiple of the de Broglie wavelength associated with the electron revolving around the orbit.

24. What transition in the hydrogen spectrum would have the same wavelength as the Balmer transition $n = 4$ to $n = 2$ of He^+ spectrum?

25. Calculate the energy required for the process He^+ (g) $\rightarrow$ He^{2+} (g) + e^- The ionization energy for the H atom in the ground state is 2.18×10^{-18} J atom^{-1} .

26. If the diameter of a carbon atom is 0.15 nm, calculate the number of carbon atoms which can be placed side by side in a straight line across length of scale of length 20 cm long.

27. 2×10^8 atoms of carbon are arranged side by side. Calculate the radius of carbon atom if the length of this arrangement is 2.4 cm.

28. The diameter of zinc atom is 2.6 Å. Calculate (a) radius of zinc atom in pm and (b) number of atoms present in a length of 1.6 cm if the zinc atoms are arranged side by side lengthwise.

29. A certain particle carries 2.5×10^{-16}C of static electric charge. Calculate the number of electrons present in it.

30. In Milikan's experiment, static electric charge on the oil drops has been obtained by shining X-rays. If the static electric charge on the oil drop is -1.282×10^{-18}C, calculate the number of electrons present on it.

31. In Rutherford's experiment, generally the thin foil of heavy atoms, like gold, platinum etc. have been used to be bombarded by the α-particles. If the thin foil of light atoms like aluminium etc. is used, what difference would be observed from the above results ?

32. Calculate the wavelength for the emission transition if it starts from the orbit having radius 1.3225 nm and ends at 211.6 pm. Name the series to which this transition belongs and the region of the spectrum.

33. Dual behaviour of matter proposed by de Broglie led to the discovery of electron microscope often used for the highly magnified images of biological molecules and other type of material. If the velocity of the electron in this microscope is 1.6×10^6 ms^{-1}, calculate de Broglie wavelength associated with this electron.

34. Similar to electron diffraction, neutron diffraction microscope is also used for the determination of the structure of molecules. If the wavelength used here is 800 pm, calculate the characteristic velocity associated with the neutron.

35. If the velocity of the electron in Bohr's first orbit is 2.19×10^6 ms^{-1}, calculate the de Broglie wavelength associated with it.

36. The velocity associated with a proton moving in a potential difference of 1000 V is 4.37×10^5 ms^{-1}. If the hockey ball of mass 0.1 kg is moving with this velocity, calculate the wavelength associated with this velocity.